

Cross-cohort contrast vectors identify an ISS-T-aligned matrix-high/DCT-low kidney remodeling state in mouse spaceflight transcriptomes

RRRM-2 contrast analysis with OSD-513 recurrence, OSD-253 strain context, and
live-return arm attenuation/divergence

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Independent research draft for mentor feedback
Authorship/collaboration structure pending

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Executive summary for mentor feedback

Purpose of this draft. This document is being shared to ask whether the claim hierarchy is biologically and statistically defensible, and whether the project is worth developing toward a preprint or journal submission.

Question	Can RRRM-2 kidney RNA-seq support an age- and arm-dependent spaceflight network-remodeling claim, and what biologically meaningful arm-specific signal is supported across OSDR cohorts?
Datasets	RRRM-2 / OSD-771 is the factorial bridge cohort. OSD-513 is the primary external recurrence cohort. OSD-253 is used as a metadata-aware mechanism/context cohort, especially for C57BL/6J versus C3H/HeJ and original versus white-LED rerun ground controls. Tabula Muris Senis kidney is used as an external aging reference.
Original claim status	RRRM-2 cannot support the original age-rewiring claim at primary module/pathway scale. Within-RRRM-2 control-aging vectors are too unstable for the planned contrast-of-contrasts interpretation.
Main supported result	RRRM-2 ISS-T aligns with an independent OSD-513 kidney flight vector, while RRRM-2 LAR does not show a detectable pooled matrix-minus-DCT flight response. The aligned ISS-T direction is a matrix-high / DCT-low tubulointerstitial state. LAR diverges from ISS-T at mechanism-score scale, but the apparent young-LAR opposite-direction state effect is underpowered and not statistically distinguishable from zero. Mechanism-axis projections nominate matrix/S1P/innate, clock-DCT, and tubular-transport programs as follow-up hypotheses rather than confirmatory subdivisions.
Mechanism boundary	OSD-253 suggests TLR4/innate association, not TLR4 requirement. The remodeling score is lower in C3H/HeJ at the point-estimate level, but strain-by-flight deltas are not significant and ECM/fibrosis/adhesion/proteolysis scores remain positive under the white-LED rerun control scenario.
Feedback requested	Is this claim hierarchy biologically/statistically defensible? Is the cross-OSDR remodeling-axis story worth developing toward preprint/submission? What additional analysis, validation, or reframing would make the paper credible to space biology, kidney biology, or computational genomics reviewers?

Working claim structure. RRRM-2 cannot support the original broad age-rewiring claim. RRRM-2 ISS-T aligns with OSD-513, and the aligned direction is matrix-high and DCT/NCC-WNK-low. LAR is best described as attenuation at the pooled state-score level, with no detectable matrix-minus-DCT flight response; mechanism-score vectors suggest LAR divergence from ISS-T, and age-stratified patterns remain hypothesis-generating at $n = 5$ per cell. OSD-253 suggests TLR4/innate association, not TLR4 requirement. WGCNA and sample-specific network methods are used as annotation and prioritization layers.

Plain-language summary

The original goal was to test whether spaceflight changes how mouse kidney aging networks behave. RRRM-2 is unusually useful for this because it includes young and old mice, flight and ground controls, and two mission arms. Each condition has only five mice, so the within-cohort aging-network direction is too noisy for the original broad age-rewiring claim. The supported result is arm-specific remodeling. The ISS terminal dissection arm of RRRM-2 points in the same pathway direction as an independent kidney spaceflight cohort, OSD-513. That shared direction looks like extracellular-matrix/fibrosis activation with distal tubule transport suppression. The live animal return arm does not show the same pooled state shift. Mechanism-score vectors suggest LAR divergence from ISS-T, and the young-LAR point estimate goes in the opposite state-score direction, but that age-stratified effect is not statistically resolved. OSD-253 helps constrain mechanism: TLR4/innate biology tracks the score, while the strain comparison argues for a broader remodeling program. LIONESS and node2vec prioritize candidate local network-neighborhood contributors to that program.

Abstract

Spaceflight affects renal physiology, but the extent to which mouse kidney transcriptomes show age-dependent network remodeling or recovery-associated redirection remains unclear. We analyzed RRRM-2 (OSD-771), an age-by-arm-by-flight mouse kidney RNA-seq dataset, together with external OSDR cohorts using residualized expression, curated pathway scores, WGCNA modules, LIONESS sample-specific network summaries, node2vec local-neighborhood displacement, cross-cohort contrast vectors, and a live-return arm attenuation/divergence analysis. RRRM-2 could not support the original broad age-rewiring claim because control-aging vectors were unstable at primary module/pathway resolution. The strongest result emerged at the arm-specific flight-vector level. RRRM-2 ISS-T aligned with OSD-513 at pathway scale (cosine 0.641; 95% CI 0.351 to 0.814; directional permutation $p = 0.127$), whereas RRRM-2 LAR anti-aligned directionally with OSD-513 (cosine -0.511 ; CI -0.726 to -0.231 ; directional permutation $p = 0.178$).

The recurring ISS-T/OSD-513 direction was extracellular-matrix/fibrosis-positive and DCT/NCC-WNK transport-negative. A state-space decomposition represented this as a matrix-high / DCT-low tubulointerstitial state, with the DCT component kept in its native biological direction and the derived matrix-minus-DCT scalar kept separate from vector geometry. RRRM-2 ISS-T showed a strong positive state shift (matrix-minus-DCT effect $+2.254$, BH $q = 0.020$), whereas LAR was near zero at the pooled level (-0.112 , BH $q = 0.895$). The pooled arm-by-flight interaction was significant (2.366 , permutation $p = 0.016$, BH $q = 0.0267$), consistent with ISS-T response and LAR attenuation. Age-stratified effects were suggestive but underpowered: young ISS-T shifted strongly toward the state ($+3.204$, BH $q = 0.025$), while young LAR had an opposite point estimate (-0.782) that was not statistically distinguishable from zero (Welch $p = 0.361$, BH $q = 0.542$). Mechanism-score vectors supported LAR divergence from ISS-T (cosine -0.725 , bootstrap CI -0.874 to -0.122), and exploratory mechanism-axis projections nominate matrix/S1P/innate, clock-DCT, and tubular-transport programs for follow-up. OSD-253 constrained mechanism: the remodeling program was attenuated at the point-estimate level in C3H/HeJ but remained positive, and several ECM/fibrosis/adhesion/proteolysis scores remained positive in C3H/HeJ. WGCNA and LIONESS/node2vec nominated exploratory local-network contributors. These results support a reframed claim: RRRM-2 captures an ISS-T-aligned kidney remodeling state that recurs across cohorts, while live animal return primarily shows attenuation and mechanism-level divergence rather than a confirmed age-specific reversal.

Keywords: spaceflight transcriptomics; mouse kidney; OSD-771; RRRM-2; OSD-513; OSD-253; LIONESS; node2vec; WGCNA; extracellular matrix; renal fibrosis; TLR4; S1P; circadian clock genes; distal convoluted tubule; NCC-WNK transport; live animal return.

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1 Introduction

1.1 Spaceflight kidney biology is broader than kidney-stone risk

Kidney risk has been part of human spaceflight medicine since early discussions of weightlessness-associated urolithiasis [8]. Subsequent astronaut studies connected long-duration missions to altered calcium homeostasis, bone loss, and increased renal stone risk [21, 30]. More recent space medicine has broadened this concern from stone formation alone to kidney remodeling, tubulopathy, solute handling, mitochondrial stress, vascular and interstitial changes, and multi-organ physiology [1, 7, 29]. Rodent spaceflight studies are central to this shift because kidney tissue can be collected directly and analyzed by histology, transcriptomics, and multi-omic profiling under controlled mission conditions [9–11].

The biological question is not only whether spaceflight changes mean expression of individual genes. Kidney function depends on coordinated nephron-segment transport, immune activation, endothelial and interstitial remodeling, extracellular-matrix turnover, and stress-response programs. These processes can move together at module or pathway scale even when individual genes do not pass stringent differential-expression thresholds. Conversely, gene-level network metrics can look interesting without supporting a cohort-level biological program. The present study therefore treats network analysis as a multi-resolution problem: claims are strongest when pathway/module-scale direction, external recurrence, mechanism scoring, and exploratory gene prioritization point toward compatible biology.

1.2 RRRM-2 defines a clean factorial contrast with limited per-cell precision

NASA RRRM-2 (OSD-771) is a rare mouse kidney RNA-seq cohort with Age, mission Arm, and Environment Group represented in the same study. Young and old female C57BL/6NTac mice were sampled across flight, habitat ground-control, vivarium, and basal contexts, with ISS terminal dissection (ISS-T) and live animal return (LAR) arms. This design naturally invites a factorial question: does spaceflight amplify, dampen, or redirect age-associated kidney transcriptional network patterns differently in ISS-T and LAR?

For an arm $a \in \{\text{ISS-T}, \text{LAR}\}$, the within-arm control-aging vector and flight-aging vector can be written as

$$A_{GC}^a = \bar{X}_{\text{Old},a,GC} - \bar{X}_{\text{Young},a,GC}, \quad A_{FLT}^a = \bar{X}_{\text{Old},a,FLT} - \bar{X}_{\text{Young},a,FLT}. \quad (1)$$

The original scientific aim was to compare A_{FLT}^a with A_{GC}^a . Projection of A_{FLT}^a onto A_{GC}^a would quantify aging-like amplification or dampening, while the residual component would quantify redirection. This is a clean conceptual formulation. The statistical difficulty is that A_{GC}^a itself is estimated from only ten mice within each arm (five young GC and five old GC). A noisy control-aging direction makes downstream claims about amplification or redirection unstable.

This is a property of the estimand. High-dimensional contrast-of-contrasts are harder than simple flight-versus-ground comparisons, especially for network features whose uncertainty is already larger than mean-expression uncertainty. The analysis below therefore separates the original age-rewiring hypothesis from the cross-cohort flight-remodeling question that RRRM-2 can answer when paired with independent OSDR cohorts.

1.3 Role of WGCNA and sample-specific network analyses

Weighted gene co-expression network analysis (WGCNA) is useful for identifying co-expression modules, module eigengenes, hub genes, module-trait associations, and module preservation [16, 17]. It is a sensible structural annotation method for kidney transcriptomics. However,

standard WGCNA does not naturally test the factorial interaction at the heart of this project. A module eigengene can show age association, flight association, or arm association, but the method summarizes a module by its dominant variance axis. A smaller-variance interaction pattern within a module can be blurred by the eigengene. WGCNA is therefore valuable here as a module-scale biological layer, not as the sole primary inferential engine for age-dependent network rewiring.

Sample-specific network methods such as LIONESS estimate how each sample contributes to a shared network structure [15]. Embedding methods such as node2vec provide low-dimensional summaries of graph neighborhoods [13]. These approaches are useful for asking whether a gene’s local network context changes, in addition to whether its mean expression changes. Given the cohort size and multiple-testing burden, this manuscript uses LIONESS/node2vec as exploratory local-neighborhood prioritization after establishing the pathway/module-level recurring direction.

1.4 A cross-cohort contrast strategy

The strategy in this manuscript is to pair RRRM-2 with external OSDR cohorts at the level of comparable pathway contrasts. RRRM-2 supplies the factorial bridge: it defines ISS-T and LAR flight directions under a controlled design. OSD-513 supplies an independent flight-vector recurrence test. OSD-253 supplies mechanistic and metadata-aware context because it includes C57BL/6J and C3H/HeJ mice and because C3H/HeJ has defective TLR4 signaling, making it useful for constraining TLR4-dependent interpretations. OSD-253 is analyzed as a mechanism-context cohort because original and rerun ground-control definitions trade off light history and sequencing/read-length comparability, and rodent spaceflight transcriptomes are known to be sensitive to sample collection and preservation factors [11, 22].

The resulting hierarchy is deliberately simple. First, RRRM-2 is tested for the original age-rewiring claim. Second, RRRM-2 arm-specific flight vectors are tested against OSD-513. Third, the aligned direction is decomposed into a matrix-high / DCT-low tubulointerstitial state. Fourth, LAR is tested against three competing models: attenuation, true reversal, and mechanism switch. Fifth, OSD-253 is used to ask whether the state is TLR4-associated or TLR4-dependent. Sixth, WGCNA, LIONESS, and node2vec are used to annotate and prioritize genes within the pathway-level program.

2 Methods

2.1 Datasets and metadata handling

2.1.1 RRRM-2 / OSD-771

RRRM-2 (OSD-771) was the discovery and factorial bridge cohort. The analysis used kidney RNA-seq from female C57BL/6NTac mice in a design crossing Age (Young and Old), mission Arm (ISS-T and LAR), and Environment Group (FLT, Habitat Ground Control, Vivarium Control, Basal), with $n = 5$ biological replicates per Age $\times$ Arm $\times$ Environment cell. ISS-T and LAR were treated as distinct mission-arm/recovery-context strata rather than as a simple duration contrast. Primary flight contrasts used FLT and Habitat Ground Control samples. Vivarium and Basal samples were retained for preprocessing context and metadata completeness but were not pooled into the primary FLT-versus-GC vectors.

2.1.2 OSD-513

OSD-513 was used as the primary external recurrence cohort because it contains a powered mouse kidney flight comparison with $n = 9$ FLT and $n = 9$ Habitat Ground Control samples, plus additional context samples. It is not identical to RRRM-2 in sex, mission, strain substrain, or metadata, so it was not pooled with RRRM-2 at raw-expression level. Instead, the OSD-513 flight-minus-ground pathway vector was compared to RRRM-2 arm-specific flight vectors.

2.1.3 OSD-253

OSD-253 was used as a mechanistic and metadata-aware context cohort. It contains C57BL/6J and C3H/HeJ samples and therefore provides an opportunity to ask whether an ISS-T-aligned remodeling score is attenuated in a TLR4-impaired strain background. However, OSD-253 also has control-definition complexity. Original ground controls are closer to the original sequencing context but have light-history mismatch, whereas 2021 white-LED rerun controls improve the light comparison but introduce read-length/library comparability concerns. Therefore, both scenarios were run, and OSD-253 was not treated as a single clean validation endpoint.

2.1.4 External kidney aging reference

Tabula Muris Senis kidney data were used to construct an independent mouse kidney aging axis [33]. Female kidney single-cell profiles were aggregated to donor-level pseudobulk expression, and the primary external aging vector was defined as 18-month minus 3-month donor expression. This reference is useful because it prevents the small RRRM-2 GC aging direction from being the only definition of kidney aging, but it differs from RRRM-2 in age range, platform, and cell capture.

2.2 Expression preprocessing and residualized matrix

All primary RRRM-2 analyses used the residualized expression matrix and matched sample metadata generated by the project preprocessing pipeline. Upstream preprocessing included expression filtering, variance-stabilizing transformation, cell-type deconvolution, and residualization against technical and cellular-composition covariates. The purpose of residualization was to reduce the chance that apparent network or pathway shifts were driven only by changes in estimated kidney cell-type proportions. The residualized matrix was used for pathway scoring, module projection, and sample-specific node summaries.

External OSDR cohorts were processed with study-specific expression resources. Because OSDR studies differ in mission design, sequencing details, strain, sex, and metadata structure, external cohorts were compared at the level of within-study contrasts rather than by pooled sample-level modeling.

2.3 Feature resolutions

Four feature resolutions were used.

Pathway resolution. Curated kidney-relevant gene sets were scored per sample by averaging residualized expression across genes represented in the feature universe. The main pathway panel included DCT/NCC-WNK transport, ECM remodeling, fibrosis, apoptosis, oxidative stress, inflammation, calcium handling, lipid metabolism, and broad ion transport. DCT/NCC-WNK was designated as the primary tubular-transport feature because it is more mechanistically coherent than broad ion transport.

Module resolution. WGCNA module eigengenes from the existing module run were used as module-level features. WGCNA modules were used to annotate coherent gene programs and to

ask whether pathway-scale effects map onto unsupervised co-expression architecture.

Gene resolution. Gene-level expression features were retained for sensitivity and exploratory analyses. Gene-level analyses were never used to override module/pathway stability because they carry larger multiple-testing and interpretability burdens.

LIONESS-node and node2vec resolution. LIONESS sample-specific networks were built on a shared topology/backbone. Per-sample incident edge strengths were aggregated to gene-level node-strength features. Separately, condition-level graph embeddings were aligned by Procrustes transformation and used to compute local-neighborhood displacement. These quantities were used only for exploratory candidate prioritization.

2.4 Within-RRRM age-vector stability and decomposition

For each arm and feature resolution, a control-aging vector was computed as Old GC minus Young GC:

$$A_{GC}^a = \bar{X}_{\text{Old},a,GC} - \bar{X}_{\text{Young},a,GC}. \quad (2)$$

Bootstrap resampling was stratified by Age, Arm, and Environment Group. Each bootstrap replicate rebuilt A_{GC}^a and computed its cosine similarity to the full-sample vector. Module and pathway resolutions were treated as primary biological resolutions. Gene and LIONESS-node resolutions were treated as secondary.

If a primary control-aging direction was stable, the planned decomposition would compare the flight-aging vector

$$A_{FLT}^a = \bar{X}_{\text{Old},a,FLT} - \bar{X}_{\text{Young},a,FLT} \quad (3)$$

to A_{GC}^a . The projection coefficient

$$\beta^a = \frac{A_{FLT}^a \cdot A_{GC}^a}{\|A_{GC}^a\|^2} \quad (4)$$

would quantify alignment along the control-aging axis: $\beta > 1$ suggests amplification, $0 < \beta < 1$ suggests dampening, $\beta < 0$ suggests reversal, and a large orthogonal residual suggests redirection. In the production run, primary module/pathway stability did not pass consistently, so this decomposition is not promoted as a main biological claim.

2.5 Cross-OSDR flight-vector recurrence

The arm-specific RRRM-2 flight vector at pathway resolution was computed as age-pooled FLT minus GC within each arm:

$$F^{\text{RRRM2},a} = \bar{X}_{\text{FLT},a} - \bar{X}_{\text{GC},a}. \quad (5)$$

For an external study s , the corresponding flight vector was

$$F^s = \bar{X}_{\text{FLT}}^s - \bar{X}_{\text{GC}}^s. \quad (6)$$

Vectors were compared using cosine similarity over shared pathway features. External uncertainty was estimated by bootstrap resampling within FLT and GC strata. A label-permutation sensitivity was also added for each external study/scenario by shuffling FLT/GC labels within the external cohort while holding the RRRM-2 arm-specific reference vector fixed. Directional permutation p values report the fraction of shuffled external contrasts with cosine at least as positive as the observed value for positive alignments, or at least as negative for anti-alignments. OSD-513 was interpreted under a recurrence boundary. OSD-253 was interpreted under a context-sensitivity boundary because of metadata and control-scenario caveats.

2.6 Recurrent remodeling-axis construction

A recurrent ISS-T remodeling axis was constructed from pathways that were concordant between RRRM-2 ISS-T and OSD-513. Let u_p be the unit-normalized RRRM-2 ISS-T effect for pathway p and v_p the unit-normalized OSD-513 effect. The raw pathway weight was the mean directional contribution from the two cohorts. The final axis was oriented so ECM/fibrosis activity was positive and DCT/NCC-WNK transport suppression was negative. Pathways with discordant signs were excluded from the primary axis. Broad ion transport was retained only as a sensitivity feature when its direction was mixed.

For each sample i , a scalar remodeling score was computed as

$$S_i = \sum_{p \in P_{\text{axis}}} z(X_{ip})w_p, \quad (7)$$

where $z(X_{ip})$ is the standardized pathway activity and w_p is the unit-normalized recurrent-axis weight. Higher S_i indicates a sample that is more similar to the recurrent ISS-T/OSD-513 remodeling direction.

2.7 Mechanism scoring

Mechanism signatures were defined in the versioned mechanism gene-set configuration file. Signatures included TLR4/innate immune signaling, TGF-beta/EMT/fibrosis, ECM organization, integrin/cell adhesion, S1P/S1PR3 signaling, MMP/ADAM proteolysis, DCT/NCC-WNK transport, broad tubular transport, oxidative stress/NRF2, macrophage/inflammatory context, and preservation/sample-stress response. Signatures were assigned roles as outcome anchors, candidate upstream mechanisms, or candidate context signatures.

To reduce circularity, axis-defining genes were dropped from mechanism signatures where possible before scoring. Coverage reports recorded requested genes, matched genes, scored genes, axis-overlap removals, and dropped genes. Mechanism scores were then tested against the scalar remodeling score by Spearman correlation within RRRM-2, OSD-513, and OSD-253. RRRM-2 and OSD-513 flight-minus-ground effects were also computed for mechanism scores.

2.8 Tubulointerstitial state-space analysis

The recurrent remodeling axis was also represented as a two-dimensional tubulointerstitial state space. This step was designed to make the biology legible without sign-flipping the DCT/NCC-WNK component. For each sample, mechanism scores were standardized within each study/scenario context. The matrix component was defined as

$$M_i = \frac{z(\text{ECM}_i) + z(\text{Integrin/adhesion}_i) + z(\text{MMP/ADAM}_i)}{3}. \quad (8)$$

The DCT transport component was defined in its native direction:

$$T_i = z(\text{DCT/NCC-WNK}_i), \quad (9)$$

where higher T_i means stronger DCT/NCC-WNK transport signature. The plotted state space uses M_i on the x-axis and T_i on the y-axis. The matrix-high / DCT-low remodeling direction is therefore rightward and downward. A scalar state score was defined separately as

$$R_i = M_i - T_i. \quad (10)$$

Higher R_i means higher ECM/adhesion/proteolysis activity and lower DCT/NCC-WNK transport.

Two context components were computed in parallel: an immune-context component, defined as the mean of TLR4/innate and macrophage/inflammation z-scores, and a preservation-stress component, defined as the preservation/sample-stress z-score. FLT-minus-GC effects were computed for RRRM-2 ISS-T and LAR, age-pooled and age-stratified; OSD-513 FLT versus habitat ground control; and OSD-253 C57BL/6J and C3H/HeJ under both original-GC and white-LED rerun control definitions.

RRRM-2 state-space scores were joined to existing deconvolution outputs for DCT, fibroblast, immune, endothelial, proximal tubule, and TAL/LOH proportions. Spearman correlations tested whether M_i , T_i , immune context, preservation stress, or R_i tracked estimated cell proportions. A targeted renal gene panel was extracted from the residualized RRRM-2 expression matrix for ECM/adhesion/proteolysis genes, TLR/innate/macrophage genes, DCT/NCC-WNK transport genes, and tubular/metabolic/network-priority genes observed in earlier analyses. This panel was used for biological visualization, not as an additional genome-wide discovery test.

2.9 Live-return arm attenuation/divergence analysis

The LAR anti-alignment raised a separate biological question: does live animal return simply attenuate the ISS-T remodeling response, move in the opposite direction, or switch into another program? We therefore defined three competing models before interpreting the data. Model A was attenuation or recovery-to-baseline, where the LAR FLT-minus-GC vector is near zero relative to ISS-T. Model B was true reversal, where the LAR vector is stably opposite to ISS-T and the LAR effect itself is non-zero. Model C was mechanism switch, where LAR is not simply the negative of ISS-T but instead loads onto other programs such as clock/DCT transcriptional state, S1P/vascular-interstitial signaling, preservation/cold-stress context, immune resolution, or persistent tubular transport disruption.

For each feature set X , arm-specific flight vectors were defined as

$$v_{ISS} = \bar{X}_{ISS,FLT} - \bar{X}_{ISS,GC}, \quad v_{LAR} = \bar{X}_{LAR,FLT} - \bar{X}_{LAR,GC}. \quad (11)$$

Age-pooled effects were computed as the mean of young and old age-stratified FLT-minus-GC contrasts to keep age strata balanced. Reversal was quantified by

$$\cos(v_{LAR}, v_{ISS}) \quad (12)$$

and by the projection coefficient

$$\beta_{LAR \rightarrow ISS} = \frac{v_{LAR} \cdot v_{ISS}}{\|v_{ISS}\|^2}. \quad (13)$$

Negative cosine indicates an opposite direction; negative β indicates movement opposite the ISS-T direction with magnitude interpretable on the ISS-T scale. Bootstrap confidence intervals resampled samples within Arm $\times$ Age $\times$ condition cells. Permutation sensitivities shuffled FLT/GC labels within Arm $\times$ Age strata. For state-space vector geometry, `matrix_minus_dct` was excluded from the feature vector because it is exactly `matrix_component` minus `dct_transport_component`; it was retained only as a scalar readout in effect and interaction tables.

The reversal test was run at state-space, mechanism-score, WGCNA-module eigengene, all-gene residual-expression, and exploratory LIONESS/node2vec-summary levels. A feature set was labeled as a reversal candidate when the LAR/ISS-T cosine was negative with bootstrap CI excluding zero and $\beta_{LAR \rightarrow ISS} \leq -0.25$. A feature set was labeled attenuation when LAR magnitude was small relative to ISS-T and $|\beta| < 0.25$. Intermediate negative cosines were labeled partial reversal or switch.

Component-level arm-by-flight interactions were computed for matrix component, DCT transport, immune context, preservation stress, matrix-minus-DCT, ECM organization, S1P/S1PR3, clock-gene scores, Rbm3/preservation scores, and related targeted panels. For each feature f , the interaction was

$$\Delta_f = (\text{FLT} - \text{GC})_{\text{ISS}} - (\text{FLT} - \text{GC})_{\text{LAR}}. \quad (14)$$

Bootstrap intervals resampled within Arm $\times$ Age $\times$ condition cells, and permutation p values shuffled FLT/GC labels within Arm $\times$ Age strata. BH correction was applied across the component-interaction family. This component-interaction test is equivalent in estimand to a standard linear-model interaction on a gene-set score and is interpreted as differential gene-set activity rather than a network-specific discovery test.

Mechanism-context projections placed ISS-T and LAR effects onto hand-defined axes in mechanism-score space: matrix-high/DCT-low, clock-DCT state, S1P/matrix, preservation/cold-stress, immune/TLR4/macrophage, and transport rebound. These projections are descriptive and carry no independent confirmatory claim without component-level support. The clock-DCT axis included core clock genes (*Arntl/Bmal1*, *Clock*, *Npas2*, *Per1*, *Per2*, *Per3*, *Cry1*, *Cry2*, *Nr1d1*, *Nr1d2*, *Rora*, *Rorc*, *Dbp*, *Tef*, *Hlf*) and DCT/WNK context genes. Without time-of-day metadata, this was interpreted as a clock-gene-associated transcriptional state, not as circadian phase. S1P analysis included sphingosine metabolism and receptor genes, with *S1pr3* emphasized because it was a WGCNA hub. Rbm3 analysis evaluated *Rbm3* as both candidate biology and preservation/cold-stress context marker.

2.10 OSD-253 TLR4 discriminator

OSD-253 was analyzed under two control scenarios: original ground controls and 2021 white-LED rerun ground controls. For each scenario and score, bootstrap estimates were computed for the C57BL/6J flight effect, the C3H/HeJ flight effect, and the C3H-minus-C57 strain delta. Interpretive labels were assigned as follows. A score was labeled **tlr4_supported** only if the C57BL/6J effect was positive, the C3H/HeJ effect was attenuated, and the strain delta supported attenuation. A score was labeled **tlr4_associated** if the score tracked the remodeling axis but remained positive in C3H/HeJ without a strain delta. Otherwise, it was labeled **inconclusive**. These labels summarize the strain-discriminator result for downstream interpretation.

2.11 Exploratory LIONESS/node2vec local-neighborhood prioritization

For each gene g , node2vec local-neighborhood displacement was computed separately in ISS-T and LAR:

$$D_g^{\text{ISS}} = 1 - \cos(v_g^{\text{ISS,FLT}}, v_g^{\text{ISS,GC}}), \quad D_g^{\text{LAR}} = 1 - \cos(v_g^{\text{LAR,FLT}}, v_g^{\text{LAR,GC}}). \quad (15)$$

ISS-T-specific embedding displacement was

$$\text{Spec}_g = z(D_g^{\text{ISS}}) - z(D_g^{\text{LAR}}). \quad (16)$$

LIONESS node-strength effects were computed as FLT-minus-GC node-strength differences within each arm:

$$N_g^{\text{ISS}} = \bar{N}_{g,\text{ISS,FLT}} - \bar{N}_{g,\text{ISS,GC}}, \quad N_g^{\text{LAR}} = \bar{N}_{g,\text{LAR,FLT}} - \bar{N}_{g,\text{LAR,GC}}. \quad (17)$$

The exploratory composite score was

$$C_g = z(D_g^{\text{ISS}}) - z(D_g^{\text{LAR}}) + z(|N_g^{\text{ISS}}|) - z(|N_g^{\text{LAR}}|) + 0.5z(\text{OSD513 support}_g). \quad (18)$$

An optional silent-shifter score penalized genes with strong ISS-T differential expression:

$$C_g^{\text{silent}} = C_g - z(|\log FC_g^{\text{ISS}}|). \quad (19)$$

These scores nominate local network-neighborhood contributors to the recurrent remodeling program and are reported as prioritization metrics rather than formal gene-level discovery tests.

2.12 Gene-priority enrichment and sensitivity

Reported top-k enrichment used proportional thresholds only: top 1%, 2%, 5%, and 10% of the phase-specific gene universe. These correspond to 25, 50, 125, and 250 genes in the 2,500-gene production universe, and 50, 100, 250, and 500 genes in the 5,000-gene sensitivity universe. Fisher exact tests used the phase-specific gene universe as background and were BH-corrected across mechanism signatures within each ranking-score/threshold family. Ranked-list enrichment used a Mann-Whitney approach comparing composite scores inside versus outside each mechanism signature, with BH correction across signatures within each ranking score.

2.13 Multiple testing and uncertainty accounting

Benjamini-Hochberg correction was applied within explicitly defined test families [2]. For mechanism-score correlations, FDR correction was applied across all non-missing score-versus-remodeling Spearman tests across the three study/scenario contexts reported in the mechanism association table. For OSD-513 mechanism flight effects, FDR correction was applied across the mechanism signatures tested in the OSD-513 FLT-versus-GC contrast. For the LAR attenuation/divergence analysis, vector-level permutation statistics were reported with BH-adjusted values across the vector-summary table, but these q-values are treated as descriptive because feature sets and age scopes are nested and overlapping. Component-interaction permutation tests were corrected across the tested state/mechanism/targeted components and interpreted as gene-set-score interaction tests. Age-stratified arm-specific component effects were reported as descriptive FLT-minus-GC estimates unless explicitly tied to an interaction statistic. For the OSD-253 strain discriminator, FDR correction was applied separately within each control scenario and within each endpoint family: C57BL/6J flight effect, C3H/HeJ flight effect, and C3H-minus-C57 strain delta. For the exploratory gene-priority layer, top-k Fisher tests were corrected separately for each ranking score and proportional top-k threshold across mechanism signatures; ranked-list tests were corrected separately for each ranking score across mechanism signatures. WGCNA module-overlap tests were corrected across tested modules. Bootstrap confidence intervals for vector alignment, stability, and OSD-253 score effects were interpreted as uncertainty intervals rather than multiple-testing-adjusted significance tests. Cross-OSDR label-permutation p values were reported as contrast-level sensitivity statistics and were not pooled into the mechanism-score FDR families.

The bootstrap resampled biological samples within fixed analysis definitions. It therefore captures sampling variability under the chosen preprocessing, residualization model, gene sets, pathway panel, and cohort metadata rules. It does not propagate uncertainty from gene-set curation, cell-type deconvolution reference choice, residualization model selection, feature-universe choice, or pathway-selection decisions. Sensitivity analyses addressed part of this fixed-pipeline uncertainty by using both OSD-253 control scenarios, a 2,500-gene production network universe and 5,000-gene sensitivity universe, proportional top-k thresholds, and leave-one-pathway-out cosine diagnostics. These sensitivity analyses do not replace prospective validation.

2.14 Statistical software and reproducibility

Analyses were implemented as a reproducible Python/R pipeline with versioned configuration files, run manifests, and scripted figure generation. Differential-expression and RNA-seq preprocessing resources follow standard Bioconductor practices [19, 24, 25]. Repository-internal commands, run identifiers, and output paths are listed in the Data and code availability section rather than in the main Methods text.

3 Results

3.1 RRRM-2 cannot resolve the original age-rewiring contrast at primary resolution

RRRM-2 was designed well for a factorial question, but its per-cell sample size is small for estimating high-dimensional aging directions. Bootstrap stability of within-arm GC aging vectors showed that primary module/pathway control-aging directions were unstable in three of four arm-resolution settings (Table 1). LAR module passed, and gene/LIONESS-node summaries passed as secondary resolutions.

Arm	Resolution	Full norm	Median cosine	Lower CI	Required lower	Passed
ISS-T	pathway	0.167	0.787	0.248	0.400	no
ISS-T	module	0.325	0.750	0.056	0.400	no
ISS-T	gene	10.152	0.746	0.545	0.200	yes
ISS-T	LIONESS-node	10.152	0.745	0.549	0.200	yes
LAR	pathway	0.221	0.890	0.345	0.400	no
LAR	module	0.353	0.774	0.493	0.400	yes
LAR	gene	9.687	0.742	0.563	0.200	yes
LAR	LIONESS-node	9.687	0.737	0.556	0.200	yes

Table 1: **Bootstrap angular stability of RRRM-2 control-aging vectors.** Module and pathway are the primary biological resolutions. Gene and LIONESS-node are secondary exploratory resolutions.

This result sets the scale of the manuscript. RRRM-2 provides an arm-stratified flight contrast that can be tested for recurrence in external OSDR cohorts. The secondary stability of gene and LIONESS-node features motivates exploratory local-neighborhood prioritization, while the pathway/module claim rests on cross-cohort flight-vector recurrence.

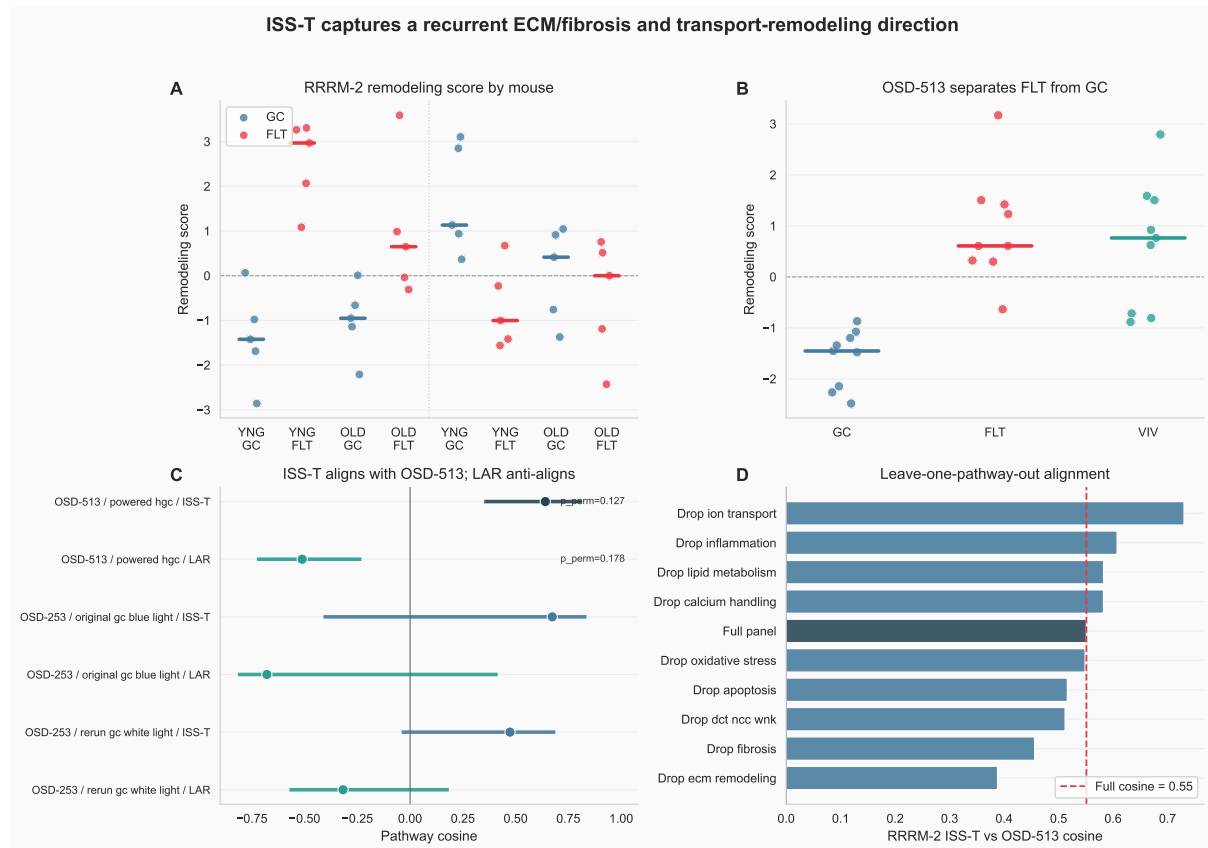


Figure 1: Main result overview. (A) RRRM-2 mouse-level remodeling scores by arm, age, and flight group. Points are individual mice; colored bars are group medians. (B) OSD-513 mouse-level separation on the same remodeling score. (C) Cross-OSDR pathway-vector alignment shows ISS-T alignment with OSD-513 and LAR anti-alignment. (D) Leave-one-pathway-out analysis shows that the alignment is not driven by a single pathway alone.

3.2 ISS-T aligns with OSD-513; LAR anti-aligns

The strongest positive result was arm-specific cross-OSDR alignment at pathway resolution. The RRRM-2 ISS-T flight vector aligned with the OSD-513 flight vector with cosine 0.641 and bootstrap 95% CI 0.351 to 0.814. In the label-permutation sensitivity, the OSD-513 ISS-T cosine was above the null median near zero but did not reach a 0.05 directional permutation threshold ($p_{\text{perm}} = 0.127$). The RRRM-2 LAR vector was anti-aligned with OSD-513, with cosine -0.511 , CI -0.726 to -0.231 , and directional permutation $p = 0.178$ (Table 2; Figure 2). OSD-253 C57BL/6J comparisons had positive ISS-T point estimates under both control definitions but wide confidence intervals crossing zero, so OSD-253 was carried forward as mechanism/context rather than recurrence evidence.

Study	Scenario	Arm	Point	Median	Lower CI	Upper CI	p_{perm}	Boundary	Claimable
OSD-513	powered HGC	ISS-T	0.641	0.625	0.351	0.814	0.127	recurrence	yes
OSD-513	powered HGC	LAR	-0.511	-0.498	-0.726	-0.231	0.178	recurrence	no
OSD-253	original GC blue light	ISS-T	0.674	0.602	-0.411	0.836	0.141	context	no
OSD-253	original GC blue light	LAR	-0.679	-0.608	-0.816	0.415	0.125	context	no
OSD-253	rerun GC white light	ISS-T	0.474	0.470	-0.040	0.688	0.269	context	no
OSD-253	rerun GC white light	LAR	-0.318	-0.309	-0.572	0.183	0.344	context	no

Table 2: **Cross-OSDR pathway-vector alignment.** OSD-513 is interpreted as directional recurrence. OSD-253 is interpreted as context sensitivity because of control-scenario caveats. p_{perm} is the directional external-label permutation sensitivity with the RRRM-2 reference vector held fixed.

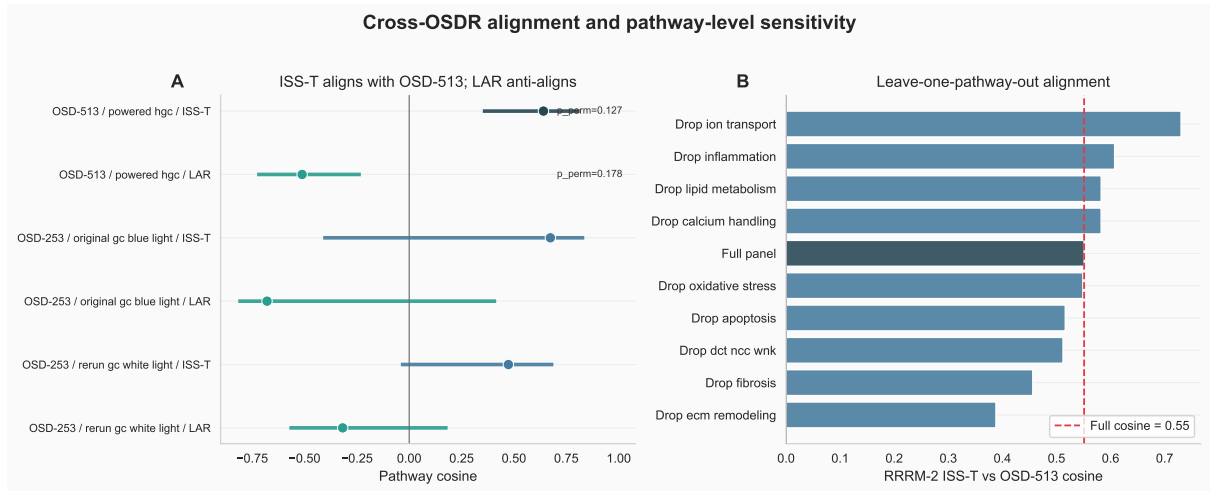


Figure 2: **Cross-OSDR alignment and leave-one-pathway-out sensitivity.** (A) RRRM-2 arm-specific flight vectors compared with external OSDR pathway vectors. ISS-T aligns with OSD-513, whereas LAR anti-aligns; OSD-513 labels show directional external-label permutation p values. (B) Leave-one-pathway-out cosines show that the RRRM-2 ISS-T/OSD-513 alignment is distributed across the pathway panel rather than fully dependent on a single component.

The arm-specific sign pattern is the key biological finding. The terminal-flight arm captures a recurring flight-associated kidney remodeling direction. The return/recovery arm does not recapitulate that direction and is directionally opposite to OSD-513 in the pathway-vector comparison, especially for ECM/fibrosis features. This distinction is consistent with live-return attenuation or context-dependent divergence, while still leaving recovery timing and collection context as linked explanations.

3.3 The recurrent axis is ECM/fibrosis-positive and DCT/NCC-WNK-negative

Pathway-level decomposition identified the features that define the recurring ISS-T/OSD-513 direction. Concordant positive pathways were ECM remodeling, fibrosis, apoptosis, and oxidative stress. DCT/NCC-WNK transport was concordantly negative and therefore received a negative weight. Calcium handling, inflammation, broad ion transport, and lipid metabolism were excluded from the primary axis because they were discordant or mixed. Broad ion transport was retained as sensitivity only. The sample-level pathway scores show the same biological direction: ECM organization increases and DCT/NCC-WNK transport decreases most clearly in the RRRM-2 ISS-T and OSD-513 flight comparisons (Figure 3).

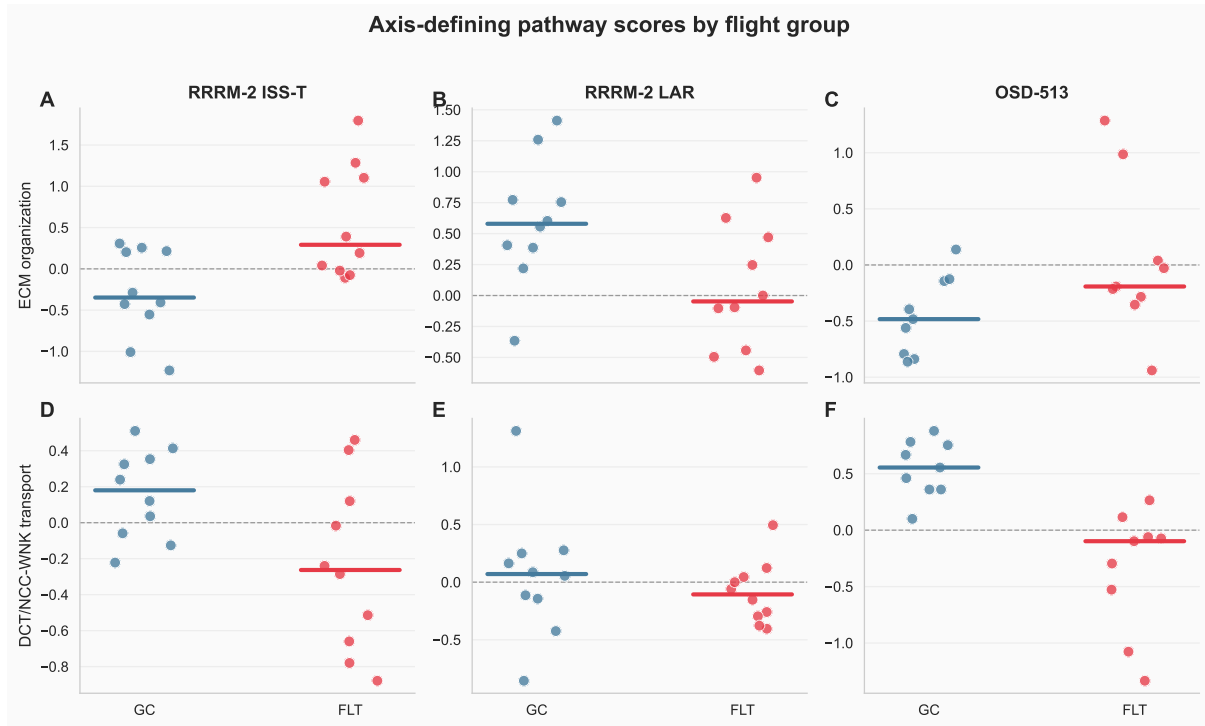


Figure 3: **Axis-defining pathway scores by flight group.** Points are individual mice, and colored bars show medians. ECM organization increases and DCT/NCC-WNK transport decreases in the aligned RRRM-2 ISS-T and OSD-513 flight contexts.

Pathway	RRRM-2 ISS-T effect	OSD-513 effect	Axis weight	Primary axis
ECM remodeling	0.949	0.801	0.622	yes
Fibrosis	0.764	0.652	0.504	yes
DCT/NCC-WNK	-0.398	-0.889	-0.452	yes
Apoptosis	0.674	0.386	0.378	yes
Oxidative stress	0.198	0.104	0.108	yes
Inflammation	0.561	-0.092	0.000	no
Ion transport	0.329	-0.762	0.000	sensitivity
Lipid metabolism	-0.320	0.156	0.000	no
Calcium handling	0.003	-0.550	0.000	no

Table 3: **Pathway construction of the recurrent ISS-T remodeling axis.** Effects are pathway-scale FLT-minus-GC contrasts. The final axis is oriented so ECM/fibrosis is positive and DCT/NCC-WNK suppression is negative.

This axis provides the biological language for the rest of the manuscript: an ECM/fibrosis-positive and DCT/NCC-WNK-negative remodeling score that recurs between RRRM-2 ISS-T and OSD-513.

3.4 A matrix-high / DCT-low tubulointerstitial state emerges across cohort context

The recurrent axis can be read more directly as a two-dimensional tissue-state shift. The matrix component summarizes ECM organization, integrin/cell adhesion, and MMP/ADAM proteolysis. The DCT transport component is kept in its native biological direction, so higher values mean stronger DCT/NCC-WNK transport. The remodeling direction is therefore rightward

and downward in state space, and the scalar `matrix_minus_dct` is defined as matrix component minus DCT transport component (Figure 4).

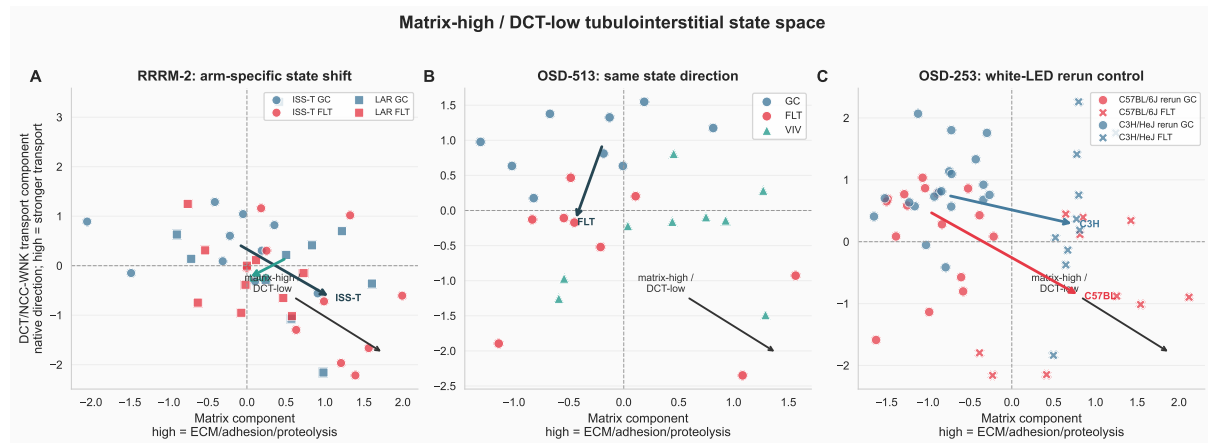


Figure 4: Matrix-high / DCT-low tubulointerstitial state space. The x-axis is the matrix component, where higher values indicate higher ECM/adhesion/proteolysis activity. The y-axis is the DCT/NCC-WNK transport component in native direction, where higher values indicate stronger DCT transport signature. The remodeling direction is rightward and downward. Points are individual mice. Arrows connect group medians from control to flight within the plotted context.

The state-space effects sharpen the biological interpretation (Figure 5). RRRM-2 ISS-T showed a positive matrix shift (FLT-minus-GC effect = 1.25, $q = 0.0158$), a negative DCT transport shift (-1.00 , $q = 0.102$), and a strong positive matrix-minus-DCT state shift (2.25 , $q = 0.0163$). Immune-context and preservation-stress components were also higher in ISS-T flight (1.16 , $q = 0.0012$; 1.06 , $q = 0.0310$). RRRM-2 LAR did not show the same state shift: matrix-minus-DCT was near zero (-0.11 , $q = 0.904$).

OSD-513 was directionally consistent at the scalar state level, with matrix-minus-DCT increasing in flight (1.81 , $q = 0.0310$). This OSD-513 state shift was driven most strongly by DCT/NCC-WNK suppression (-1.57 , $q = 0.0082$), while the matrix component was positive but not significant (0.25 , $q = 0.627$). This distinction matters: OSD-513 supports the same matrix-high/DCT-low direction, but it does not independently reproduce every component with equal strength. The OSD-513 label-permutation caveat remains part of the interpretation.

Under the OSD-253 white-LED rerun control scenario, both strains moved toward the matrix-high/DCT-low state. C57BL/6J showed a strong scalar effect (2.68 , $q = 2.3 \times 10^{-4}$) with matrix increase (1.64 , $q = 8.7 \times 10^{-4}$) and DCT decrease (-1.04 , $q = 0.072$). C3H/HeJ also showed a positive scalar effect (1.82 , $q = 0.0031$) and matrix increase (1.51 , $q = 3.1 \times 10^{-6}$), while its DCT component was only weakly negative (-0.31 , $q = 0.612$). Under the original-GC scenario, the scalar effects were weaker (C57BL/6J 0.79 , $q = 0.229$; C3H/HeJ -0.29 , $q = 0.633$), reinforcing that OSD-253 is sensitive to control definition.

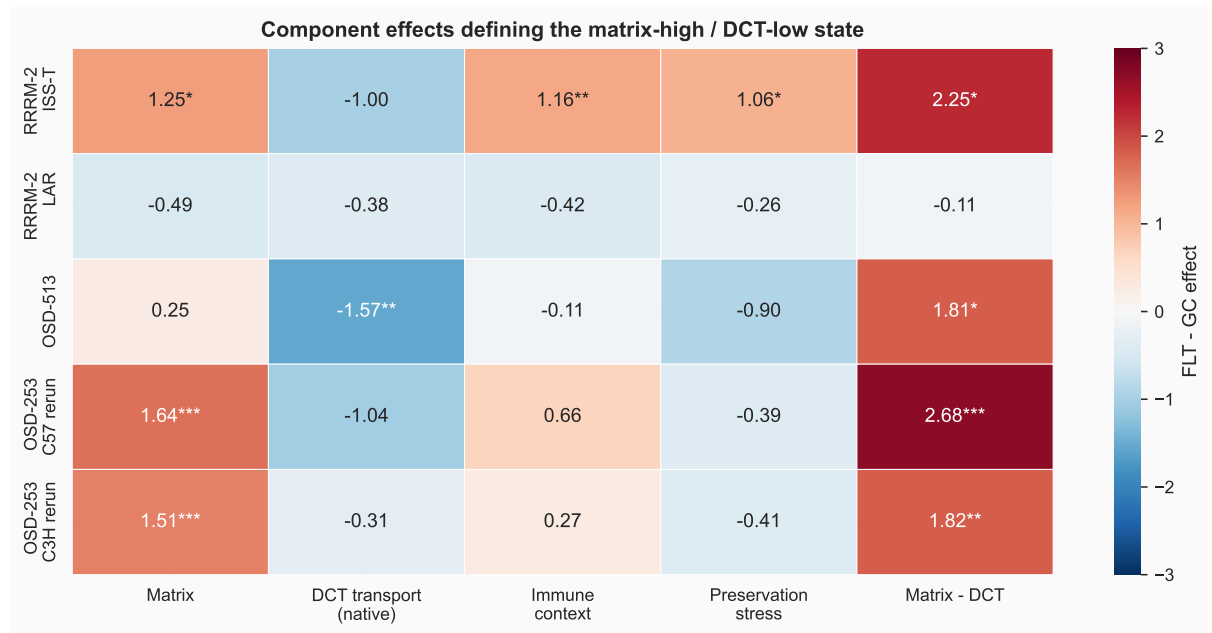


Figure 5: **Component effects defining the matrix-high / DCT-low state.** Values are FLT-minus-GC effects in standardized component units. Asterisks indicate BH-adjusted thresholds within the component-effect table. DCT transport is shown in native direction, so negative values indicate lower DCT/NCC-WNK transport signature in flight.

RRRM-2 deconvolution estimates did not materially track the state components (Figure 6). All 80 RRRM-2 samples joined to the deconvolution table. Across tested state-component/cell-proportion pairs, the largest absolute Spearman correlation was small ($|\rho| = 0.120$ for matrix component versus CLR fibroblast estimate), and all BH-adjusted q values were approximately 0.99. Within the available deconvolution model, the matrix-high/DCT-low state is therefore not simply a readout of estimated DCT or fibroblast fraction.



Figure 6: **RRRM-2 deconvolution overlay for the tubulointerstitial state.** Points are individual RRRM-2 mice. The scalar state score and its components show little correlation with estimated DCT or fibroblast proportions, suggesting that the state-space signal is not explained by these deconvolution estimates alone.

Finally, targeted residualized-expression visualization placed familiar renal genes inside the state interpretation (Figure 7). The panel includes 53 scored genes across ECM/adhesion/proteolysis, TLR/innate/macrophage, DCT/NCC-WNK transport, and tubular/metabolic-priority classes. This is a biological annotation display rather than an additional discovery test, but it is useful for seeing that the state-space result is carried by recognizable renal-remodeling and tubular-transport genes.

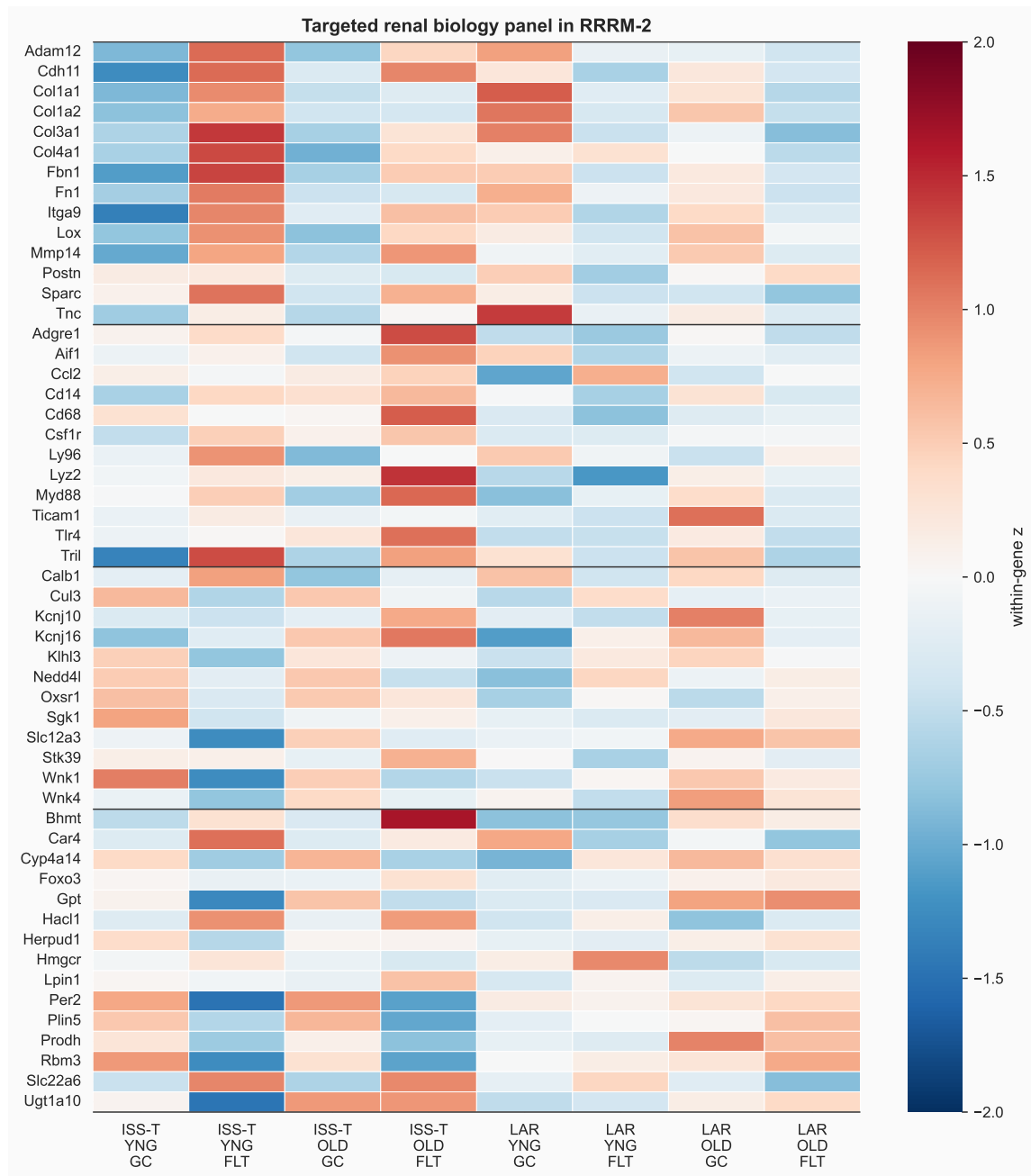


Figure 7: **Targeted renal biology panel in RRRM-2.** Values are group-mean, within-gene z-scores from residualized expression. Rows are grouped by biology class: ECM/adhesion/proteolysis, TLR/innate/macrophage, DCT/NCC-WNK transport, and tubular/metabolic-priority genes.

3.5 LAR shows attenuation at state-score level and divergence at mechanism-score level

The LAR arm was next tested against three competing models: attenuation/recovery toward baseline, true reversal of the ISS-T state, and mechanism switch into a different program. After removing the derived `matrix_minus_dct` scalar from the state-space vector geometry, the pooled LAR state-space vector remained weakly opposite to ISS-T but was unstable (cosine = -0.564 , bootstrap CI -0.940 to $+0.339$, $\beta_{\text{LAR} \rightarrow \text{ISS}} = -0.199$). This supports attenuation rather than a stable state-space reversal. Young LAR had an extreme negative point-estimate cosine (cosine

= -0.982 , directional permutation $p = 0.0030$, BH $q = 0.0450$), but its bootstrap CI remained wide and crossed zero (-0.969 to $+0.559$), so the young state-space direction is not a robust estimate.

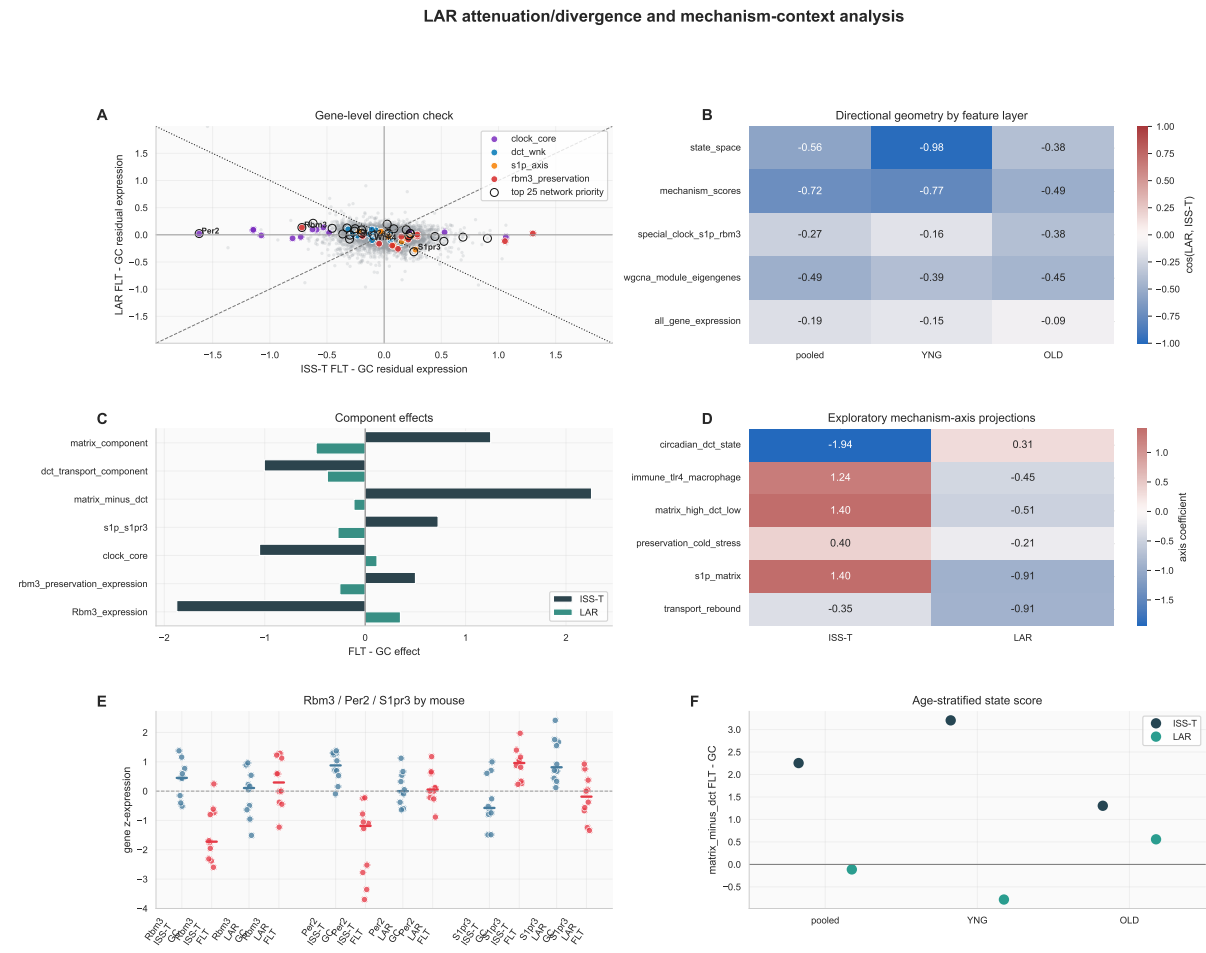


Figure 8: LAR attenuation/divergence and mechanism-context dashboard. Panel A shows RRRM-2 ISS-T and LAR gene-level FLT-minus-GC effects, with curated renal biology classes highlighted. Panel B summarizes cosine alignment across feature sets. Panel C shows arm-specific component effects. Panel D projects ISS-T and LAR effects onto exploratory mechanism-context axes. Panel E shows targeted *Rbm3*, *Per2*, and *S1pr3* expression effects. Panel F shows age-stratified matrix-minus-DCT state effects. The visually prominent young-LAR opposite point estimate is interpreted as hypothesis-generating because the LAR effect itself is not statistically distinguishable from zero.

The scalar state-score effects are most consistent with ISS-T response and LAR attenuation. Young ISS-T moved strongly toward the matrix-high/DCT-low state (matrix-minus-DCT effect = $+3.204$, BH $q = 0.025$), while young LAR had an opposite point estimate (-0.782) that was not statistically distinguishable from zero (Welch $p = 0.361$, BH $q = 0.542$). Old ISS-T showed a weaker positive shift ($+1.304$), and old LAR was weakly positive ($+0.558$). At the pooled level, ISS-T showed a strong state shift ($+2.254$, BH $q = 0.020$), whereas LAR was approximately zero (-0.112 , BH $q = 0.895$). The pooled arm-by-flight interaction for the matrix-minus-DCT state was positive and significant (interaction = 2.366 , bootstrap CI 0.865 to 3.851 , permutation $p = 0.0160$, BH $q = 0.0267$), but this interaction is driven by ISS-T response against an attenuated LAR response, not by a confirmed non-zero LAR reversal.

Component interactions identify which biology differs most between arms. Matrix was the most robust component: ISS-T increased and LAR decreased the matrix component at the point-estimate level (interaction = 1.738, BH $q = 0.0037$), although the LAR matrix effect itself had a bootstrap interval crossing zero. DCT transport was lower in ISS-T and also lower in pooled LAR, and the pooled DCT interaction was not significant (BH $q = 0.346$). Thus, LAR does not show clear DCT recovery.

Mechanism-score vectors provided the strongest evidence that LAR diverges from ISS-T beyond simple state-score attenuation. Pooled mechanism-score vectors showed LAR opposite to ISS-T (cosine = -0.725 , bootstrap CI -0.874 to -0.122 , directional permutation $p = 0.0160$, BH $q = 0.0800$). The clearest component-level divergences involved ECM organization, TLR4/innate signaling, S1P/S1PR3, and *S1pr3* expression. S1P/S1PR3 increased in ISS-T and decreased in LAR (interaction = 0.997, BH $q = 0.0027$), and *S1pr3* expression showed strong arm divergence (ISS-T effect = $+1.276$, LAR effect = -1.269 , interaction = 2.545, BH $q = 0.0020$).

Clock-associated DCT features nominate a second follow-up axis. Core clock score decreased in ISS-T but was slightly positive in LAR (interaction = -1.170 , BH $q = 0.0020$); the Per/Cry subscore showed the same arm divergence (interaction = -2.061 , BH $q = 0.0024$). *Per2* decreased strongly in ISS-T and was approximately unchanged in LAR (interaction = -2.563 , BH $q = 0.0024$). Across RRRM-2 samples, *Per2* expression correlated positively with the DCT transport component ($\rho = 0.449$, BH $q = 7.5 \times 10^{-5}$) and negatively with matrix-minus-DCT ($\rho = -0.547$, BH $q = 7.6 \times 10^{-7}$). Because RRRM-2 does not provide a time-series design or sufficient time-of-day metadata for phase inference, this result is described as a clock-gene-associated transcriptional state, not a measured circadian phase shift.

Exploratory mechanism-axis projections were retained as context, not as confirmatory decomposition. The hand-defined axes explained 37% of the ISS-T mechanism-feature variance and 51% of the LAR variance, leaving substantial residual signal. Their coefficients are therefore useful as biological sketches: matrix/S1P/innate features point toward ISS-T-positive/LAR-negative biology, clock-DCT features suggest ISS-T suppression with LAR attenuation or partial rebound, and transport-rebound features remain negative in both arms. These projections nominate mechanisms to test, but they do not establish three separable programs as a primary result.

At broader feature levels, WGCNA module eigengenes also pointed oppositely (pooled cosine = -0.489 , bootstrap CI -0.763 to -0.080), though permutation support was weaker. All-gene expression was much weaker (cosine = -0.187), arguing against a genome-wide reversal. Node2vec displacement magnitudes were moved to a separate context table because magnitude vectors are non-directional and not comparable to signed effect vectors. Therefore, the LAR result is best framed as pooled attenuation with mechanism-level divergence, plus an underpowered age-stratified observation that may motivate validation.

3.6 Mechanism scores nominate ECM, innate, adhesion, S1P, and DCT programs

Mechanism scoring asked which candidate upstream or contextual programs track the scalar remodeling score. In RRRM-2, the remodeling score correlated strongly with ECM organization ($\rho = 0.644$, $q = 7.4 \times 10^{-10}$), DCT/NCC-WNK transport suppression ($\rho = -0.560$, $q = 3.1 \times 10^{-7}$), TLR4/innate signaling ($\rho = 0.488$, $q = 1.8 \times 10^{-5}$), MMP/ADAM proteolysis ($\rho = 0.478$, $q = 2.6 \times 10^{-5}$), S1P/S1PR3 signaling ($\rho = 0.451$, $q = 8.2 \times 10^{-5}$), and integrin/cell adhesion ($\rho = 0.374$, $q = 1.5 \times 10^{-3}$). OSD-513 reproduced the ECM-positive and DCT-negative relationship: ECM organization correlated with the remodeling score ($\rho = 0.745$, $q = 2.7 \times 10^{-5}$), and DCT/NCC-WNK transport correlated negatively ($\rho = -0.676$, $q = 2.8 \times 10^{-4}$). OSD-253, across all samples, showed strong associations of the remodeling score with TLR4/innate

signaling, ECM organization, macrophage/inflammatory context, DCT/NCC-WNK suppression, and S1P/S1PR3 (Figure 9).

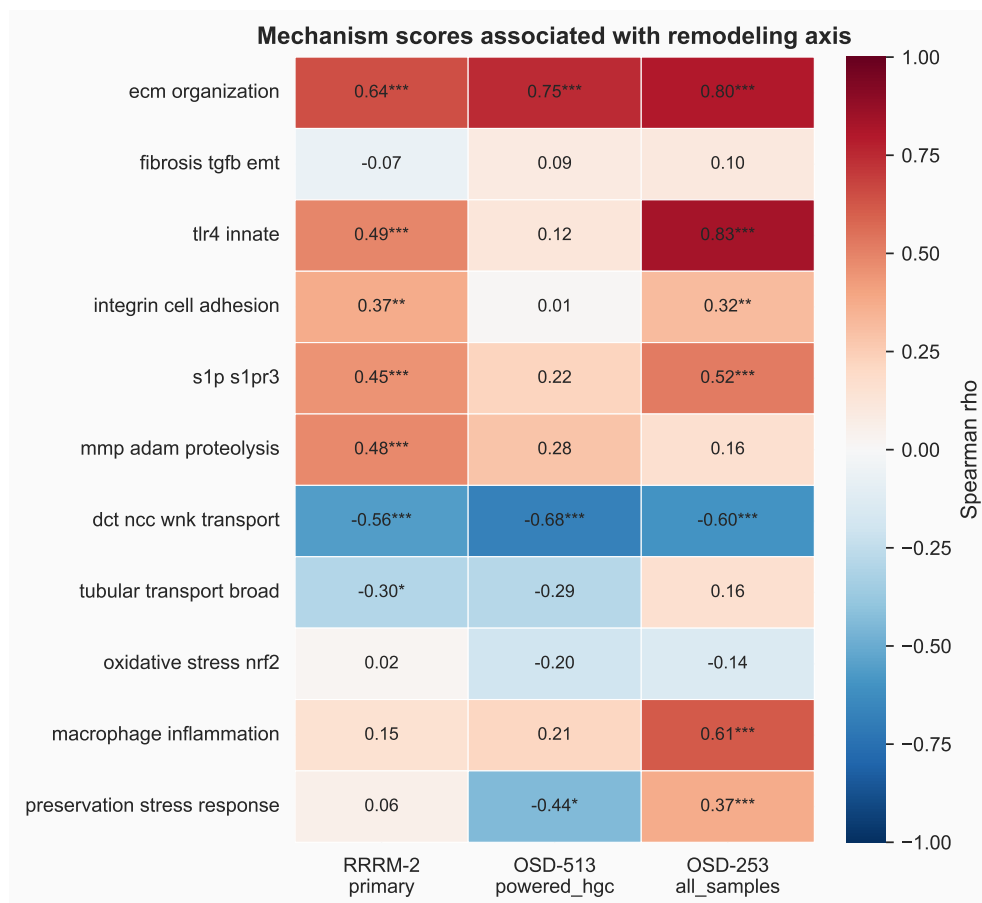


Figure 9: **Mechanism-score associations with the recurrent remodeling score.** Color indicates Spearman correlation between each mechanism score and the scalar remodeling score. Asterisks mark BH-adjusted $q < 0.05$ within the association table. RRRM-2 and OSD-513 both show ECM-positive and DCT/NCC-WNK-negative structure. OSD-253 shows strong TLR4/innate, ECM, macrophage/inflammation, and transport relationships across its broader sample set.

The mechanism pattern is biologically plausible. TLR4 signaling is implicated in renal inflammation and fibrosis, and TLR4-mutant mice have been reported to be protected from chronic kidney disease progression and renal fibrosis [32]. TRIL is a co-receptor associated with TLR3/TLR4 signaling and cytokine production [3, 4, 34]. Cadherin-11 has a recognized role in tissue fibrosis and fibroblast-like cell adhesion [6]. Integrins and MMP/ADAM proteases are core mediators of cell-matrix adhesion and matrix turnover. Sphingosine-1-phosphate signaling, including S1PR3-centered biology, has been linked to kidney injury, inflammation, and fibrotic remodeling in several contexts [27, 28]. The DCT/NCC-WNK direction is also coherent with distal nephron transport regulation, where WNK-SPAK/OSR1-NCC signaling governs sodium-chloride cotransport and potassium-sensitive distal nephron physiology [5, 12].

The mechanism scores place the recurrent remodeling direction in a molecular neighborhood involving ECM deposition, adhesion/proteolysis, innate immune signaling, S1P biology, macrophage/inflammatory context, and tubular transport suppression. This supports a mechanism-prioritization model in which multiple upstream and downstream programs are carried forward for targeted validation.

3.7 OSD-253 points to TLR4/innate association rather than TLR4 requirement

OSD-253 was used as the immediate mechanistic discriminator because C3H/HeJ mice have impaired TLR4 signaling. A strict TLR4-dependent model predicts attenuation of the remodeling score in C3H/HeJ relative to C57BL/6J. The observed pattern supports an association model (Figure 10). Under the original GC blue-light scenario, C57BL/6J remodeling was positive (0.816) and C3H/HeJ remodeling was lower (0.328), with a strain-delta confidence interval crossing zero. Under the white-LED rerun scenario, remodeling was positive in both strains (C57BL/6J 1.379, C3H/HeJ 0.668), again with a strain-delta confidence interval crossing zero. ECM, fibrosis/TGF-beta, integrin/cell adhesion, and MMP/ADAM scores remained positive in C3H/HeJ under the rerun scenario.

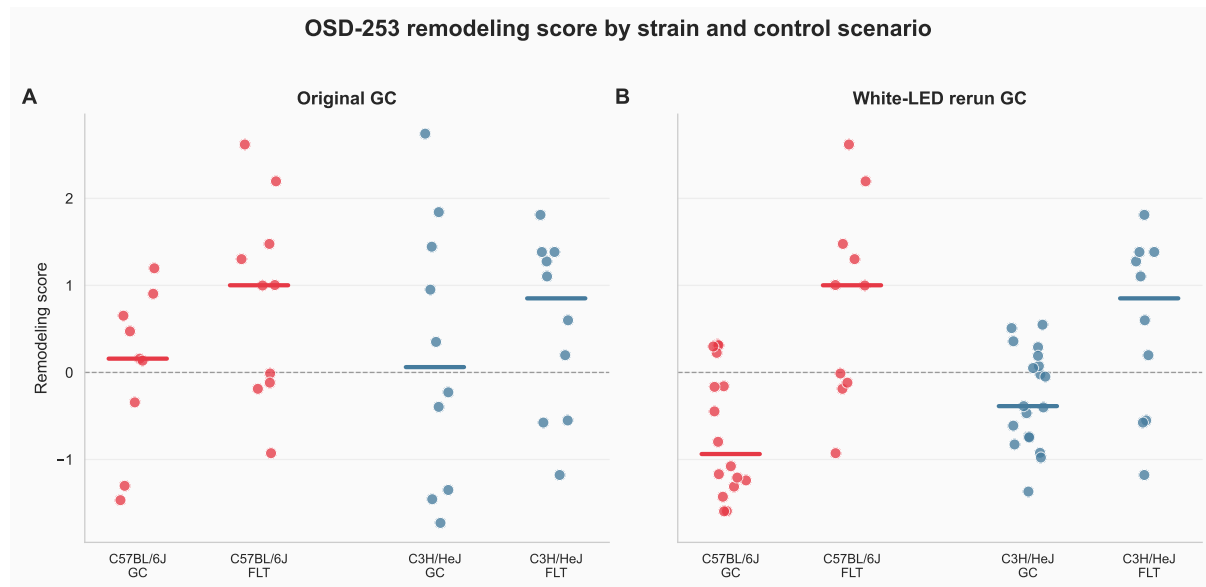


Figure 10: **OSD-253 remodeling score by strain and control scenario.** Points are individual mice, and colored bars show medians. The remodeling score is lower in C3H/HeJ at the point-estimate level, but the program remains positive under the white-LED rerun control scenario.

TLR4/innate signaling tracks the remodeling score strongly, especially in OSD-253, and remains a plausible contextual axis. The strain comparison favors a TLR4-associated but broader remodeling program. Other mechanisms, including TGF-beta/EMT, adhesion, proteolysis, S1P signaling, macrophage context, and preservation stress, remain plausible contributors or covariates.

3.8 WGCNA provides structural annotation of the remodeling program

The WGCNA module layer annotates the remodeling program. A prior WGCNA analysis identified grey60 as a 48-gene ECM/cell-migration module with an ISS-T-specific eigengene pattern and hub structure involving genes such as *Tril*, *Itga9*, *Fbn1*, *S1pr3*, *Cdh11*, *Adam12*, and *Mmp14*. That hub composition is consistent with the mechanism-axis result.

Follow-up overlap analyses place grey60 inside a broader multi-module response. The 11 ISS-T-Young LIONESS factorial edge-rewiring candidates distributed across purple, turquoise, yellow, black, brown, and one off-panel feature. Silent-shifter candidates enriched midnightblue in ISS-T-Young (11/185; $q = 0.0031$) and purple in ISS-T-Old (13/171; $q = 0.030$). Module eigengene

contrasts showed an ISS-T-specific pattern in grey60, royalblue, salmon, and pink. WGCNA therefore supports structural annotation of a multi-module remodeling response, with grey60 as the most biologically interpretable ECM/cell-migration module.

3.9 LIONESS/node2vec prioritizes candidate local-neighborhood contributors

The exploratory LIONESS/node2vec ranking asked which genes show ISS-T-specific local network-neighborhood movement, stronger ISS-T node-strength effects than LAR, and support from OSD-513. In the 2,500-gene production run, the top composite genes included *Rbm3*, *Cyp4a14*, *Bhmt*, *Per2*, *Hmgcr*, *Sim2*, *Hacl1*, *Plin5*, *Gpt*, *Prodh*, and *Ugt1a10*. In the 5,000-gene sensitivity run, high-ranking genes included *Rbm3*, *Bhmt*, *Kank3*, *Ugt1a10*, *Ptpn1*, *Serpib8*, *Cd200*, *Per2*, and *Cyp4a14*.

These genes are candidate local-neighborhood contributors to the pathway-level program. Top-k mechanism enrichment was weak after FDR correction (Figure 11). In the 2,500-gene production run, the strongest top-k findings were broad tubular transport at top 250 (2/4 genes; fold enrichment 5.0; raw $p = 0.052$; $q = 0.384$) and preservation stress at top 250 (3/10; fold enrichment 3.0; raw $p = 0.070$; $q = 0.384$). Ranked-list enrichment suggested broad tubular transport enrichment in the composite score (raw $p = 0.0085$, $q = 0.076$). In the 5,000-gene sensitivity run, the best ranked-list signals were preservation stress, broad tubular transport, and oxidative stress, all with $q = 0.130$ or weaker.

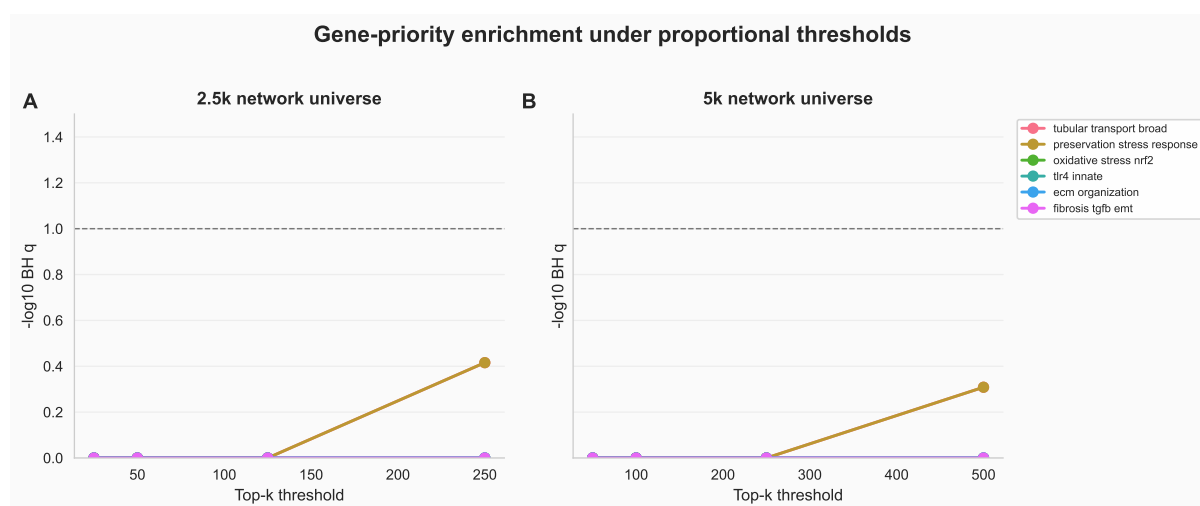


Figure 11: **Top-k sensitivity for LIONESS/node2vec gene prioritization.** The gene-priority layer nominates candidate local-neighborhood contributors. The most consistent weak signals involve broad tubular transport, preservation stress, and oxidative/NRF2 context, while FDR support remains below the pathway/module-scale evidence.

Node2vec and LIONESS add a gene-prioritization layer: they rank genes whose local network context moves with the ISS-T remodeling program. This layer is useful for follow-up target selection and gene-level annotation of the pathway/module result.

3.10 External aging-axis projection favors remodeling over aging mimicry

Projection onto the Tabula Muris Senis kidney aging axis separated the ISS-T remodeling program from simple aging mimicry. ISS-T was essentially orthogonal to the external aging vector ($\beta = -0.0027$, cosine = -0.0187). LAR had a small positive projection ($\beta = 0.0162$,

cosine = 0.175), but the magnitude was small and the residual component remained dominant. The ISS-T signal is therefore best described as recurrent remodeling and transport biology rather than accelerated kidney aging.

3.11 Claim hierarchy emerging from the full analysis

The final claim hierarchy is:

1. RRRM-2 does not resolve the planned within-cohort age-dependent rewiring contrast at primary module/pathway scale.
2. RRRM-2 ISS-T aligns with an independent OSD-513 kidney flight direction; RRRM-2 LAR directionally anti-aligns but does not pass the external-label permutation null.
3. The aligned ISS-T direction is a matrix-high / DCT-low tubulointerstitial state.
4. LAR shows no detectable pooled matrix-minus-DCT flight response, so the primary LAR interpretation is attenuation relative to ISS-T. The young-LAR opposite point estimate is hypothesis-generating because the LAR effect itself is not significant and the state-space bootstrap CI crosses zero.
5. Mechanism-score vectors support LAR divergence from ISS-T. Matrix/S1P/innate, clock-DCT, and tubular-transport projections are retained as follow-up mechanism hypotheses, not as confirmed separable programs.
6. OSD-253 suggests TLR4/innate association, while the C3H/HeJ comparison favors a broader remodeling program rather than strict TLR4 requirement.
7. WGCNA, LIONESS, and node2vec provide structural annotation and exploratory gene prioritization below the cross-cohort pathway/module and mechanism-state results.

4 Discussion

4.1 Principal biological interpretation

The central biological result of this study is not that different mission arms can differ. That is expected from the design. The result is the specific renal state that appears when kidney tissue is captured at terminal flight. Across RRRM-2 ISS-T and OSD-513, flight is associated with a coordinated matrix-high / DCT-low tubulointerstitial state: extracellular-matrix organization, cell adhesion, proteolysis, innate/stress-associated biology, and S1P/S1PR3-related features rise, while distal convoluted tubule transport programs, especially DCT/NCC-WNK-linked biology, fall. This state is the strongest coherent biological object recovered by the analysis.

This interpretation reframes the project away from a broad age-rewiring model. RRRM-2 was initially attractive because its factorial design could, in principle, ask whether spaceflight amplifies, dampens, or redirects age-associated kidney networks. In practice, the within-cohort control-aging directions were too unstable at primary module/pathway resolution to support that claim. The more defensible biology is a flight-associated tubulointerstitial remodeling program that recurs across independent kidney spaceflight cohorts and is modulated by live return.

4.2 A tubulointerstitial remodeling state

The word remodeling is used deliberately. The transcriptomic pattern is not simply generic differential expression, and it is not by itself proof of established fibrosis. It is a tissue-state signal

in which matrix, adhesion, proteolysis, innate immune context, and tubular transport move together. In kidney biology, these processes are plausibly linked: renal injury and repair programs commonly involve epithelial stress, inflammatory signaling, fibroblast and matrix activation, altered cell-matrix adhesion, and changes in segment-specific transport identity [6, 14, 18, 20]. The present data place the spaceflight-associated kidney response in that injury-repair/remodeling neighborhood.

The distal tubule side of the axis is especially important. DCT/NCC-WNK biology is not only a marker panel; it represents a physiologic transport system involved in sodium-chloride handling, potassium-sensitive WNK-SPAK/OSR1 signaling, and downstream electrolyte and blood-pressure regulation [5, 12]. Suppression of this program alongside ECM/adhesion/proteolysis activity suggests a kidney state in which specialized distal nephron transport identity is reduced while matrix and interstitial remodeling programs are engaged. That coupling is the main biological signal in the study.

This does not mean that transcriptomics has measured fibrosis, transport function, or renal injury directly. The appropriate conclusion is more constrained: terminal-flight kidney transcriptomes carry molecular evidence for a matrix-remodeling/high-stress state coupled to lower distal-nephron transport signatures. The result nominates concrete tissue and physiology endpoints for validation, including collagen deposition, tubular injury markers, immune infiltration, vascular/interstitial markers, urine and serum chemistry, and segment-specific NCC/WNK abundance or phosphorylation.

4.3 What live return changes

The live animal return arm is biologically informative because it does not simply preserve the terminal-flight state. At the pooled scalar level, LAR shows no detectable matrix-minus-DCT state shift, whereas ISS-T shows a clear positive shift. This supports attenuation of the terminal-flight remodeling state after return rather than a confirmed reversal. The age-stratified young-LAR opposite point estimate remains hypothesis-generating because the LAR effect is not statistically distinguishable from zero.

The more interesting LAR interpretation is decoupling. Matrix/S1P/innate features and *S1pr3* show strong arm divergence, with ISS-T-positive and LAR-negative behavior. DCT transport features, however, do not show a clean pooled recovery. Thus, return may reduce or redirect the matrix/innate/S1P arm of the terminal-flight response while leaving distal transport biology incompletely recovered, weakly affected, or differently organized. In biological terms, the data are more consistent with a transition between an active terminal-flight injury-repair state and a post-return recovery/reloading state than with a simple “flight effect disappears” model.

Clock-associated DCT features add a plausible but limited recovery-context signal. *Per2* and clock-score behavior separate ISS-T from LAR and correlate with the DCT/state components. Because RRRM-2 is not a time-course experiment and time-of-day information is insufficient for phase inference, this should be described as a clock-gene-associated transcriptional state, not as evidence of circadian phase shifting.

4.4 Mechanistic neighborhood

The mechanism-score results place the remodeling state near ECM organization, adhesion/proteolysis, TLR4/innate signaling, macrophage/inflammatory context, S1P/S1PR3 biology, and DCT/NCC-WNK suppression. This is a plausible renal injury-remodeling neighborhood. TLR4 and innate immune signaling can promote inflammatory and fibrotic kidney injury, TRIL is connected to TLR3/TLR4 signaling, integrins and cadherins participate in cell-matrix and fibroblast-like adhesion, MMP/ADAM proteases mediate matrix turnover, and

S1P/S1PR3 signaling has recognized links to vascular, immune, inflammatory, and fibrotic responses [3, 4, 26–28, 32, 34].

OSD-253 helps set a boundary around this interpretation. If the state were strictly TLR4-dependent, it should collapse in the TLR4-impaired C3H/HeJ context. Instead, the remodeling score is lower in C3H/HeJ at the point-estimate level, but strain-delta confidence intervals cross zero, and ECM/fibrosis/adhesion/proteolysis scores remain positive in C3H/HeJ under the white-LED rerun control scenario. The safer conclusion is that TLR4/innate biology is associated with, and may amplify or contextualize, the remodeling state; it is not established as a required single upstream driver.

4.5 Why this is not accelerated kidney aging

The negative aging result is important. Projection onto an external Tabula Muris Senis kidney aging axis did not support a simple accelerated-aging interpretation, and RRRM-2 control-aging vectors were too unstable to support the original age-rewiring claim. This does not make the analysis uninformative. It narrows the biology. The ISS-T signal is better described as flight-associated renal remodeling with distal transport suppression than as aged kidney mimicry. Age may still modify susceptibility or recovery, but the current design cannot make that interaction a primary claim.

4.6 Role of WGCNA, LIONESS, and node2vec

The network methods are best understood as annotation and prioritization layers below the pathway/state result. WGCNA maps the remodeling state onto co-expression structure, especially an ECM/cell-migration module containing genes such as *Tril*, *Itga9*, *Fbn1*, *S1pr3*, *Cdh11*, *Adam12*, and *Mmp14*. That module structure gives the pathway-level result a recognizable gene neighborhood.

LIONESS and node2vec answer a different, secondary question: which genes may have local network neighborhoods that move with the ISS-T remodeling program? The top-ranked genes include stress-response, metabolic, clock-associated, transport, and signaling candidates, including *Rbm3*, *Per2*, *Cyp4a14*, and *Bhmt*. These methods do not establish a separate gene-level topology-reversal claim. Their value is to nominate candidate genes and local neighborhoods for follow-up inside the broader remodeling/transport state.

4.7 Limitations

Several limitations constrain the interpretation. First, RRRM-2 has only $n = 5$ per Age $\times$ Arm $\times$ Environment cell, limiting high-dimensional interaction estimation and making age-stratified LAR patterns descriptive. Second, ISS-T and LAR differ in flight state, return/recovery timing, reloading, and collection context; their contrast should not be interpreted as a pure recovery experiment. Third, the clock-DCT result cannot be interpreted as circadian phase because RRRM-2 is not a circadian time-course experiment. Fourth, external OSDR cohorts differ in sex, strain, mission, duration, sequencing, and metadata. Cross-cohort agreement at pathway-vector level is therefore meaningful, but raw expression pooling would be inappropriate. Fifth, OSD-253 has two imperfect control definitions, so it should remain a mechanism-context cohort rather than a clean validation endpoint. Sixth, curated mechanism signatures and bootstrap procedures cannot propagate every source of pipeline-level uncertainty.

4.8 Future work

The most direct follow-up is tissue-level validation of the remodeling/transport axis. Collagen staining, tubular injury markers, immune and macrophage markers, vascular/interstitial

markers, and segment-specific NCC/WNK assays would test whether the transcriptomic state corresponds to structural remodeling, tubular injury, or altered distal-nephron transport machinery. The LAR-specific question is whether return and reloading preferentially attenuate matrix/S1P/innate activity while leaving distal transport programs incompletely restored. A design powered for Age $\times$ Arm $\times$ Flight interaction would be needed before age-specific recovery or reversal can be claimed.

Mechanistically, TLR4-intact and TLR4-impaired comparisons should be paired with S1P/S1PR3, TGF-beta/ECM, adhesion/proteolysis, macrophage/innate, and DCT/WNK markers. Computationally, the next useful step is not to add more network machinery, but to preserve the claim hierarchy: cross-cohort pathway/state recurrence first, mechanism-context interpretation second, and network-neighborhood gene prioritization third.

5 Conclusion

RRRM-2's strongest estimable claim is a terminal-flight renal remodeling state that recurs in OSD-513. The state is matrix-high and DCT/NCC-WNK-low, placing the kidney response near injury-repair, ECM/adhesion/proteolysis, innate/S1P-associated signaling, and distal transport suppression rather than simple accelerated aging. Live animal return attenuates or redirects the matrix-associated component of this state more clearly than it restores distal transport biology. OSD-253 supports TLR4/innate association within a broader remodeling program, not strict TLR4 requirement. WGCNA, LIONESS, and node2vec provide structural annotation and candidate-gene prioritization below this primary biological claim.

Author contributions

I.S. designed and implemented the analytical framework, conducted the contrast-vector, cross-OSDR, mechanism-axis, WGCNA-context, LIONESS/node2vec, and sensitivity analyses, generated the figures, drafted the manuscript, and maintained the source code, tests, and run artifacts. Authorship and collaboration structure will be revised with mentor and collaborator input.

Competing interests

The author declares no competing interests.

Data and code availability

NASA Open Science Data Repository accessions: OSD-771 (RRRM-2), OSD-513, and OSD-253. Tabula Muris Senis kidney was used as the external aging reference. Pipeline source, configuration files, run manifests, tests, figures, and manuscript drafts are version-controlled in the project repository.

- Main v6 production run: `run_20260519_000547_2500g`.
- Cross-OSDR permutation sensitivity run: `run_20260519_cosine_perm_null`.
- Primary mechanism-axis outputs: `data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis`.
- Tubulointerstitial state-space outputs: `data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state`.

- Live-return attenuation/divergence outputs: `data/results/run_20260519_000547_2500g/contrast_vectors/mechanism_axis/tubulointerstitial_state/lar_reversal`.
- 5,000-gene sensitivity run: `run_20260519_5000g_sensitivity_3seed`.
- Figure-generation scripts: `scripts/plot_manuscript_v4_figures.py`; `scripts/plot_lar_reversal_dashboard.py`.
- Pipeline entry point: `venv/bin/python-msrc.run_all_phases--updated-pipeline`.

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